INCH-POUND

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MILITARY SPECIFICATION

PUNCHING MACHINES, METAL, SINGLE END TYPE, COMPUTER NUMERICALLY CONTROLLED (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers computer numerically controlled (CNC), metal punching machines of the single end type, with equipment and accessories specified herein.

1.2 Classification. The machines shall be of the following classes. The sizes shall be as listed in table I. The class, and size machine to be furnished shall be as specified (see 6.2.1).

Class A - Single punch

Class B - Multi - Punch (Tool Changer)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

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AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers Association, Standards Department, Suite 201, 1500 King Street, Arlington, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA) STANDARDS

RS-227 One Inch Perforated Paper Tape.

RS-232 Interface Between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Interchange.

RS-244 Character Codes for Numerical Machine Tool Control Perforated Tape.

RS-267 Axis and Motion Nomenclature for Numerically Controlled Machine Tools.

RS-274 Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines.

RS-281 Electrical and Construction Standards for Numerical Machine Control. (DOD) adopted)

RS-358 Subset of American National Standard Codes for Information Interchanged for Numerical Machine Control Perforated Tape.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W. Washington DC 20006-1807.

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JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete punching machine for first article inspection and approval (see 4.4 and 6.3).

3.2 Design. The punching machine shall be new (not a prototype) and one of the manufacturer's current production models, single end type CNC, and capable of metal punching operation in accordance with the requirements herein. The machine shall be an electrically powered, motor driven unit with a solid state soft-wired, integrated circuit CNC system. The machine shall be supported by a base and frame designed with a workpiece opening or throat. The machine shall be provided with tool holders for securing the punches and dies in the punching position. Class B machine shall be provided with a tool changing mechanism designed to select and position a punch, die and stripper assembly in the hole punching position under the ram of the machine drive. Tool changer rotation, workpiece positioning and the punching action of the machine shall be controlled by the CNC system. The design shall provide convenient,

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efficient, and safe operation. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Design, construction, and system of measurement. Unless otherwise required, either the inch-pound system or the International System of Units (SI) shall be used in the design and construction of the punching machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in inch-pound units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise required, either the inch-pound or the SI system of measurements shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. Unless otherwise specified (see 6.2.1), measuring and indicating devices shall be graduated to read in both the inch-pound and the SI system of measurements.

3.2.2 Reclaimed materials. The punching machine may contain reclaimed materials provided such materials will not jeopardize the intended use, performance, and design life of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The punching machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the punching machine that present safety hazards. The safety devices shall not interfere with the operation of the punching machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the punching machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 that are applicable to the machine itself. The punching machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial punching machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because

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the guard configuration is dependent on the size and configuration of the workpieces. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The punching machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the punching machine.

3.2.8 Environmental protection. The punching machine shall be so designed and constructed that, under the operation, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. An automatic lubrication system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The punching machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.5 Painting. Unless otherwise specified (see 6.2.1), the punching machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts, either inch-pound or SI, used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 390.03 for the English inch-pound system or ISO 54 for the metric (SI) system. The gears, shafts, and pinions shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shaft, and pinions in the drive train shall be hardened steel. Hardness shall be not less than Rockwsell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The machine shall consist of not less than a frame, table, drive system, tool holders, strippers, punches and dies, tool changer, electrical system, motors and CNC system.

3.4.1 Frame. The frame shall be a stress relieved construction of either iron or steel. Unless otherwise specified, the frame shall be of the "C", "J", or bridge style, having a base, column, and overhang to form a workpiece opening or throat. When only one style frame is acceptable, the particular style required shall be as specified (see 6.2.1). The frame shall be sufficiently rigid to withstand stresses imposed during performance of the punching requirements specified herein. The base of the frame shall have provisions for leveling and securing the machine to the floor or foundation. When specified (see 6.2.1), vibration dampening leveling pads shall be provided. An eye or attachment points shall be provided on the frame for lifting the machine. The eye or lift points shall be capable of supporting twice the weight of the machine and its location shall not cause distortion to the frame when the machine is suspended.

3.4.2 Table. The machine shall be provided with a table for supporting the workpiece in the punching operation. The table shall be constructed of iron or steel with the top surface either hardened or provided with anti-friction bearings or rollers to facilitate workpiece travel across the table by the workpiece positioning unit. Table motion shall be accomplished on anti-friction bearings. The table motion and positioning shall be obtained by using anti-backlash ball screw and nuts or rack and pinion drive. Table axial/movement shall be accomplished by servo drives with alternating current (AC) or direct current (DC) motors of the totally enclosed permanent magnet type. The servo motors shall have sufficient power and torque to position the table at the speed required in table I. The table shall be of the size required for fully supporting any workpiece size of the capacity of the

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TABLE I - Sizes and characteristics.

CHARACTERISTICS	"J" or "C" STYLE SIZE						BRIDGE STYLE SIZE				
	1	2	3	4	5	6	7	8	9	10	11
Throat depth inches	18	40	51	51	48	60					
Workpiece positioning unit range (without repositioning the workpiece within the workpiece positioning unit):											
In "X" axis, inches	30	78	80	98	48	90	40	90	160	160	120
In "Y" axis, inches	30	40	51	51	87	60	40	60	84	120	90
Workpiece positioning Speed along the "X" and "Y" axis, inches per minute	700	1800	1600	1800	600	1200	2200	1100	1100	1100	500
Hole punching capacity in SAE 1020 Steel: maximum diameter. inches	1.0	3.0	3.0	3.0	5.0	5.0	4.125	5.0	5.0	5.0	5.0
maximum thickness inches	.075	.120	.135	.135	.125	.250	.062	.052	.250	.250	.250
Diameter, inches	1.0	1.0	1.0	1.0	1.0	.813	1.250	.813	.813	.813	2.0
maximum material thickness, inches	.125	.250	.250	250	.5	.625	.250	.625	.625	.625	1.125
Holes per minute spaced on one inch centers in the maximum material thickness	75	220	175	220	105	105	500	55	155	75	15
Punched hole centerline dimension accuracy inches	±.002	±.002	±.002	±.002	±.005	±.005	±.005	±.005	±.005	±.010	±.015
Repeatability of hole location	±.002	±.004	±.004	±.004	±.005	±.005	±.005	±.005	±.005	±.010	±.010
Workpiece size, inches	30X30	78X40	80X51	98X51	48X87	90X60	40X90	90X60	160X84	160X120	120X90

NOTE: Machine size and capacities shall be not less than the stated requirements of the size ordered. When the largest size and capacity machine is ordered the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

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machine while located at any position within the range of the workpiece positioning unit. When required, the table top dimensions shall be of the size specified (see 6.2.1).

3.4.3 Machine drive. Unless otherwise required, the punching action of the machine shall be accomplished by using either a mechanical, a hydraulic, or a combination mechanical and hydraulic type drive. When required, the type drive shall be as specified (see 6.2.1).

3.4.3.1 Mechanical type drive. The mechanical type drive shall be provided with an electric motor driven assembly of mechanical components with connecting shaft, gears, chains, belts or pulleys as required for developing and controlling the energy of a reciprocating drive ram for accomplishing punching performance specified herein. The components shall include a flywheel for supplying supplementary power to the drive motor. The flywheel shall be of cast iron or steel, and shall be dynamically balanced after machining. An eccentric or Pitman shall be provided for converting the rotary motion of the motor and flywheel to the reciprocating motion required by the ram. All rotating and sliding components shall be provided with bearings or bushings with means for positive and continuous lubrication. The energy of the drive shall be controlled by a clutch and brake arrangement that is capable of transmitting the maximum developed torque to the ram. Provisions shall be provided to automatically control the ram stroke. The ability to control the cycle by reversing under pressure shall also be provided.

3.4.3.2 Hydraulic type drive. The hydraulic type drive shall be provided with a hydraulic power unit for developing and controlling the energy of a reciprocating drive ram for accomplishing punching performance specified herein. The hydraulic power unit shall include an electric motor, hydraulic pump, fluid reservoir, ram cylinder unit, and connecting hose or tubing. The fluid reservoir shall be provided with a fluid level gage, and a hydraulic fluid filter with replaceable type element. The hydraulic cylinder unit shall be the double acting type. The drive shall be provided with hydraulic fluid control devices to include a high pressure relief valve. Connecting hose(s) or tubing shall be capable of withstanding fluid pressure up to 150 percent of the rated working pressure of the power unit. The system shall conform to the requirements of JIC H-1 hydraulic standards.

3.4.4 Tool holders. The machine shall be provided with quick change type tool holders for securing the punches and dies in the punching position under the machine ram. The tool holders shall accommodate punches and dies for punching round, square, and rectangular shaped holes of any size ranging up to and including the hole punching capacity for the type and size machine specified. Means shall be provided for radially locating and locking the punches and dies in the punching position. When the punches and dies are locked in the punching position, they shall be concentrically aligned within .002 inch.

3.4.5 Strippers. Machines that are not provided with self-stripping type punches shall be provided with strippers that prevent the workpiece from riding up with the punch when it is withdrawn from the die.

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3.4.6 Punches and dies. When required, punches and dies of the sizes and shapes specified (see 6.2.1), shall be furnished. The punches and dies shall be tool steel, hardened and ground for accuracy. The punches and dies shall be permanently marked for size and identification.

3.4.7 Tool changer. The Class B multi-punch machine shall be provided with an automatic tool changer. The machine shall be capable of automatically selecting and changing not less than 14 punch, die and strip assemblies into the position of the punching station. The selection shall be by CNC and manual control. Each tool assembly shall consist of a punch, stripper and die. Each tool assembly shall be capable of accepting all size tools within the capacity of the machine. Proper clamping of the tool assembly shall be provided. The machine shall automatically stop whenever a tool assembly is improperly clamped. The tool changer shall be capable of changing a tool assembly in 12 seconds or less. The tool changer may be located as a separate unit or mounted as an integral part of the machine.

3.4.8 Cutting unit. When required, the machine shall be provided with either a plasma arc or CO^2 laser cutting unit as specified (see 6.2.1). The unit shall be mounted on the workpiece opening or throat of the machine frame. The cutting unit on bridge style machines could be mounted on components with Y axis travel.

3.4.9 Electrical system. The electrical system shall conform to NFPA 79. Each machine shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.9.1 Primary input voltage. Unless otherwise specified (see 6.2.1), the machine shall be wired to draw all of its electrical power from a single 230/460 volt, 3-phase 60 Hz circuit. Unless otherwise specified (see 6.2.1), the machine shall be initially wired to operate from a 460 volt source.

3.4.9.2 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed, balanced, and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60 Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of NEMA MG-1.

3.4.9.3 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.9.4 Control circuit voltage. Control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage not exceeding 120 volts.

3.4.10 CNC system. The CNC shall be of the solid state, soft-wired, microprocessor, computerized, numerical control type. The system shall provide automatic control of machine functions, operating modes, punch selection and two axes control of the horizontal "X" and "Y" motion. When a plasma - arc

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or CO2 laser cutting head is furnished, the cutting head shall also be controlled by the CNC unit. The axes shall be identified in accordance with EIA standard RS-267. Unless executive program routines are provided in nonvolatile memory form (i.e. bubble memory cassette, hard disk, etc.), the controls shall be a positioning and contouring type having linear and circular interpolation and be capable of simultaneous two axis contouring. Control features shall include buffer storage, part program storage, program edit capability and control diagnostics. The control unit shall direct machine functions from part program data in memory storage, punched tape, and manual data input. The control shall initiate a halt or an error signal should a fault condition occur in the control. All necessary executive program routines shall be furnished in the form of 1 inch, eight channel, binary coded decimal punched tape for controls having volatile memory. The control unit shall conform to EIA Standard RS-281. A line voltage variation of $\pm$ 10% from normal shall not adversely affect the control system functions. The control shall be capable of functioning in ambient temperatures ranging from 50 F to 120 F and relative humidities ranging from 5% to 95%. The control shall automatically shut down when internal operating temperatures exceed safe operating temperature. Control resolution shall be not greater than .001 inch for each linear axis. The auxiliary functions of the system shall include, but not be limited to punch action, and opening and closing of the workpiece clamps. The system shall be provided with operational interlocks and functional protection to prevent the machine from operating when the punch assembly is not locked in the punching position, the machine drive brake is not released or the air or hydraulic pressure of the machine or auxiliary drive units drops below the required operating level. Interlock circuitry shall also be provided for assuring sequential operation of the system. The control unit shall be provided with a two-axis contouring and point-to-point positioning capabilities and with absolute and incremental programing. The control unit shall rotate the tool changer, if applicable, lock the programmed punch and die station in the punching position, and position the workpiece throughout the full range of travel along the "X" and "Y" axes of motion. The control unit shall automatically control the acceleration, and deceleration of all axes of motion. The control unit shall control axis movements individually, simultaneously, or in any combination. The controls unit shall control the positioning units for accomplishing any function within the capacity of the machine without loss of cycle synchronization. The smallest program increment for positioning the workpiece shall be not more than 0.001 inch and the largest shall be not less than the workpiece positioning range specified in table I.

3.4.10.1 Peripheral equipment interface. Unless otherwise specified (see 6.2.1), the control shall be equipped with an interface complying with RS-232.

3.4.10.2 Data input. The data input media shall be one inch, eight channel, punched tape in conformance with EIA Standards RS-227, RS-244, RS-274 and RS-358.

3.4.10.2.1 Tape reader. A photoelectric tape reader shall be provided. The tape reader shall be capable of holding, feeding and reading tape prepared in accordance with paragraph 3.4.10.2, at a rate of not less than 150 characters per second. The tape reader shall be supplied with standard feed and take-up spoolers that are not less than 5-1/4 inches in diameter. The tape

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reader shall be easily accessible for maintenance and tape handling. The control shall have a tape error detection capability to continously check for the following errors.

a. Odd parity per EIA RS-244
b. Even parity per EIA RS-358
c. Misaligned sprocket holes
d. No sprocket holes

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display unit to indicate fault condition.

3.4.10.2.2 Manual data input. A manual data input (MDI) unit shall be furnished to permit manual entry of alphabetical, numerical and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.10.3 Control panel. The control unit shall have an operator control panel for displaying the control functions and visual signals of the numerical control system and the machine tool. The control panel shall be provided with knobs, switches or digital input for controlling system functions and shall include at least power-on and off, cycle start-stop, emergency stop, mode selection, feed hold, feedrate override jog, jog direction, and end of block. Indicator lights or the data display unit shall indicate or display at least reading error, in-position, over--temperature, tool change if applicable, program stop, end of program, punch assembly not locked in punching position, and low air or hydraulic pressure.

3.4.10.4 Data display. Unless otherwise specified, either a cathode ray tube (CRT) or a universal alphanumeric display unit shall be provided. When only one type of data display is acceptable, the particular type shall be specified (see 6.2.1). The CRT unit shall be capable of displaying at least 250 characters simultaneously. The universal alphanumeric display unit shall be capable of being used in a variable scanning mode to interrogate all registers sequentially or held to display only a particular register as the content changes. Either unit shall display, as a minimum, the following data:

a. Sequence number readout.
b. Actual position of each axis. The readout shall at least display the accumulative command axis position of one axis at a time without loss of reference dimensions as a result of the switching.
c. Control and system error message.
d. Miscellaneous functions.

3.4.10.5 Control system modes of operation. The operational modes of the control unit shall include manual, automatic, single block, manual data input, and manual jog.

3.4.10.6 Absolute/incremental input. The control shall have absolute/incremental input capability. This mode shall be selected by a preparatory "G" code which may be switched during the tape program.

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3.4.10.7 Automatic tape code recognition. The control shall have the capability to automatically recognize and read tape punched with either EIA RS-244 or RS-358 (ASCII) character codes.

3.4.10.8 Switchable inch/metric input. The control system shall provide switchable inch-metric input capability.

3.4.10.9 Part program storage. Unless otherwise specified (see 6.2.1), the machine control unit shall be capable of storing in memory the content of not less than 150 feet of postprocessed part program tape data.

3.4.10.10 Standby battery power. Unless nonvolatile part program memory is provided, standby battery power shall be provided for the control having volatile memory to hold the size memory provided at full load for a period of not less than 72 hours when the primary power source fails.

3.4.10.11 Part program edit. A program edit capability shall be provided to allow block insertion, deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.10.12 Contouring feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that programmed.

3.4.10.13 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.10.14 Mirror image. The machine data shall be capable of producing either right or left-hand parts from same tape through the use of inversion switches for each axis. Inversion of each axis may be by keyboard entry.

3.4.10.15 Preparatory function (G-Codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274.

3.4.10.16 Miscellaneous functions (M-Codes). The miscellaneous function codes of the control shall be in accordance with EIA Standard RS-274.

3.4.10.17 Block delete. The control shall have a block delete feature to permit by-passing of blocks of programmed data.

3.4.10.18 Buffer storage. The control shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine functions. The internal storage shall delay or store multiple blocks of data until required by the controlling devices.

3.4.10.19 Block-by-block-read. The control shall have a block-by-block read function to provide for reading information one block at a time.

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3.4.10.20 Lead and reversal error compensation. Machines equipped with ball screw and nut assembly shall have lead error and reversal error compensation to permit data to be stored in the control to compensate for lead screw errors and backlash of each linear axis.

3.4.10.21 Additional features. The following additional numerical control features shall be provided:

a. Four quadrant programming.

b. Incremental and continuous jog for each axis.

c. Leading zero suppression or decimal point programming.

d. Inches per minute (IPM)/Millimeters Per Minute (MMPM), programming in all modes of operation, except point to point mode.

e. Pattern repeat.

f. Tape edit keylock.

3.4.10.22 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.10.23 Set zero. The control unit shall have a "set zero" feature to permit the specified axis register to be reset to zero establishing present axis location as zero.

3.4.10.24 Operational software. Operational software shall be furnished with the control and shall contain all logic necessary to effectively operate the machine tool. The software shall provide either resident in the control in the form of nonvolatile memory or separately in the form of one inch, eight channel, binary coded decimal punched tape for controls having volatile memory. In either case, the software shall be provided for meaningful diagnostic relating to the use of the controls, machine tool part program and operator's actions.

3.4.10.25 Positioning units. The primary command functions of the system shall be accomplished by workpiece positioning units with positioning range and speed to comply with table I. Positioning unit travel shall be obtained by electric motor driven anti-backlash ball screws and nuts or rack and pinion drive. Response of the positioning units to the primary commands for starting and stopping and the rate of motion shall be such that the specified punching accuracy is maintained. The workpiece positioning unit shall hold the workpiece by at least two clamps, which shall be adjustable for holding any workpiece size ranging up to the capacity of the machine. The workpiece positioning unit shall be provided with means to prevent the workpiece clamps from being punched during the punching operation. When specified (see 6.2.1), the workpiece positioning unit shall be provided with automatic sheet repositioner characteristics for location of sheets longer than the rated capacity of table

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I by tape command. The closed loop workpiece positioning unit shall measure and feed back to the control unit, output signals of actual workpiece position that shall be compared with the input positioning commands of the program tape data.

3.4.10.26 Postprocessor. When specified (see 6.2.1) a postprocessor shall be furnished with the capability of converting the coordinate data generated by the computer program into machine language format of the punching machine. The computer and operating software, on which the postprocessor is to operate shall be fully described under (6.2.1).

3.4.10.27 Test tape. Unless otherwise specified (see 6.2.1), a test tape and printout programmed to perform all system functions specified herein shall be furnished. The test shall operate all slides, motors, pumps, adjusting mechanisms, and accessories during each cycle of the test tape.

3.5 Sizes and characteristics. Unless otherwise specified (see 6.2.1), the size and characteristics of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. The machine shall be capable of automatic punching of holes in SAE 1020 steel to meet the capacity and accuracy requirements of table I as applicable for the machine size specified. The CNC system shall meet the performance requirements of 3.4.10.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in 3.4.4.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), all equipment normally furnished with the manufacturer's standard commercial machine shall be furnished including one set of hand tools.

3.9 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.10 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.11 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a corrosion resistant lubrication plate shall be securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

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3.11.2 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Punching capacity, maximum hole diameter
Punching capacity, maximum material thickness
Power input (volts, total amps, phase frequency)
Data of manufacture
Contract Number or Order Number
Contract Stock Number or Plant Equipment Code

3.12 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.13 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.14 Workmanship. Workmanship of the machine and accessories shall meet all requirements specified herein and shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or

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supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than one hour, during which time the machine drive, and if applicable the tool changer shall be actuated and observed to determine compliance with the requirements specified herein. The CNC system shall be operated in accordance with the manufacturer's standard operating test procedures for warm-up and run-off checks. Run-off checks of all mechanical functions of the machine and numerical control system functions shall be made in each mode of operation specified herein. Upon completion of the warm-up check procedure, the system shall be cycled continuously for a period of not less than two hours under tape control while programmed to demonstrate reliability of the equipment. There shall be no failure or malfunction of the machine or control. Should a failure or malfunction occur, the test shall be restarted from setpoint and retimed from zero hours. The program tape used for cycling shall position the machine over the entire working range and shall inlcude operation of all controlled functions of the machine. The restart of each cycle shall occur within one minute of completion. At the end of the two hour test period, the

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machine shall return to setpoint within an accuracy of plus or minus 0.005 inch. Proper operation of all controls, motors, adjustment mechanisms, and accessories shall be verified during the trail period.

4.7.2 Alignment tolerance test. Each punch and corresponding die station shall be positioned and locked in the punching position, and then checked to verify the punch and die alignment requirements of 3.4.4.

4.7.3 Performance tests. The machine shall be tested as follows to determine compliance with the performance requirements of table I as applicable for the size machine specified. When hole punching is required, SAE 1020 steel shall be used.

4.7.3.1 Positioning speed. Positioning speed shall be verified by clamping the maximum capacity size workpiece in the workpiece positioning unit through the full positioning range for the "X" and "Y" axes. The time required and distance traveled shall conform to table I.

4.7.3.2 Hole punching capacity. Hole Punching capacity shall be vertified by operating the machine for punching at least 12 holes of each of the diameters specified in table I for each of the material thicknesses specified.

4.7.3.3 Hole punching speed and accuracy. Hole punching speed and accuracy shall be verified by operating the machine to continuously punch 0.500 inch diameter holes, spaced on one inch centers, in a workpiece with a thickness of not less than the maximum material thickness capacity of the applicable size machine. The holes shall be punched for a period of one minute. Upon completion, the number of holes punched shall be counted and the resulting punching accuracy shall be measured to determine compliance with the specified requirements.

4.7.3.4 Repeatability test. Two workpieces of 1/8" thick material shall be punched in consecutive order with 24 holes each utilizing a program tape with a hole pattern of the suppliers choice. The two plates shall be inspected to verify the repeatability requirements of table I.

4.7.3.5 CNC performance test. The CNC system shall be tested by actuating control unit to operate the machine under the automatic, semi-automatic, and manual modes.

4.7.3.6 Automatic. The test tape specified in 3.4.10.27 shall be inserted in the control unit and the CNC system operated to perform the functions specified. The printout furnished shall be used to verify the specified functions.

4.7.3.7 Semi-automatic. The control unit shall be operated as specified in 4.7.3.1, but with at least six separate insertions of commands that require the machine to perform positioning functions. Each function shall be verified for accuracy.

4.7.3.8 Manual. The control unit shall be manually operated to verify each of the primary and auxiliary functions of the system.

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4.7.4 Postprocessor validation. When a postprocessor is furnished (see 3.4.10.26), the supplier is responsible for having the postprocessor validated on the user's computer to insure compatibility with the hardware and software approximately 30 days prior to deliver to the CNC machine. One or more test tapes shall be processed from the manuscript provided by the supplier to exercise all programmable functions provided in the control and to produce a test part which demonstrates all machining capabilities. Using the procedures given in MIL-STD-462.

4.7.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The machines covered by this specification are for use in producing complex punched configurations in sheet metal and plate to close tolerances by automatic control of the machine. The machine can be used to produce prototype parts from a taped program, or to produce items repetitively from a proven process to established accuracies.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, and size required (see 1.2 and table 1).

c. When first article is required for inspection and approval. (see 3.1).

d. If machine is required to be configured in a specific measurement system (inch/pound or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (inch/pound or SI), state required system (see 3.2.1.1).

f. Dual calibrations (inch/pound and SI); if different (see 3.2.1.1).

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g. Specify additional safety and health requirements (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. Electromagnetic interference control, if required (see 3.3.8).

j. If machine frame style is to be a specific type, state required style (see 3.4.1).

k. Rigging eye, if required (see 3.4.1).

l. Table top dimensions, if different (see 3.4.2)

m. If machine drive is to be a specific type, state required type (see 3.4.3).

n. Punches and dies, specify shapes and sizes required (see 3.4.6)

o. Cutting unit, if required and if a specific type, state required type (see 3.4.8)

p. Primary input power, if different (see 3.4.9.1)

q. Motor enclosure, if different (see 3.4.9.2).

r. Interface compliance with RS-232, if required (see 3.4.10.1)

s. Data display, if to be a specific type, state required type (see 3.4.10.4)

t. Part program storage, if different (see 3.4.10.9).

u. Automatic sheet repositioner, if required (see 3.4.10.25)

v. Postprocessor, if required, specify computer name, model and software required (see 3.4.10.26)

w. Test tape, if different (see 3.4.10.27).

x. Size and characteristics, if different (see 3.5).

y. Additional performance features, if required (see 3.6.1).

z. Standard equipment, if different (see 3.8).

aa. Optional equipment, if required (see 3.9).

bb. List of repair parts required (see 3.10).

cc. Lubrication plate, if different (see 3.11.1)

dd. Nameplate, if different (see 3.11.2).

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ee. Hour meter, if different (see 3.12).

ff. First Article inspection, if different (see 4.4).

gg. Quality conformance inspection, if different (see 4.5).

hh. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1)

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference of classification changes. The machine classification changes from the previous issue of this specification are as follows:

TABLE I

MIL-P-80172B		MIL-P-80172C	
Size 1	(Deleted)	Size 1	(New size)
Size 2	(Deleted)	Size 2	(New size)
Size 3	(Deleted)	Size 3	(New size)
Size 4	(Deleted)	Size 4	(New size)
Size 5	(Deleted)	Size 5	(New size)
Size 6	(Deleted)	Size 6	(New size)
Size 7	(Deleted)	Size 7	(New size)
Size 8	(Deleted)	Size 8	(New size)
Size 9	(Deleted)	Size 9	(New size)
		Size 10	(New size)
		Size 11	(New size)

6.5 Subject term (key word) listing.

Bridge Style
Computer Numerical Control (CNC)
Punching Machine
Strippers
Tool Changers

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84 - 71
DLA-GS

User activity:
Navy - MC

Preparing activity:
DLA-IP

(Project 3445-0110)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-80172C	Punching Machines, Metal, Single End Type, Computer Numerically Controlled (CNC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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